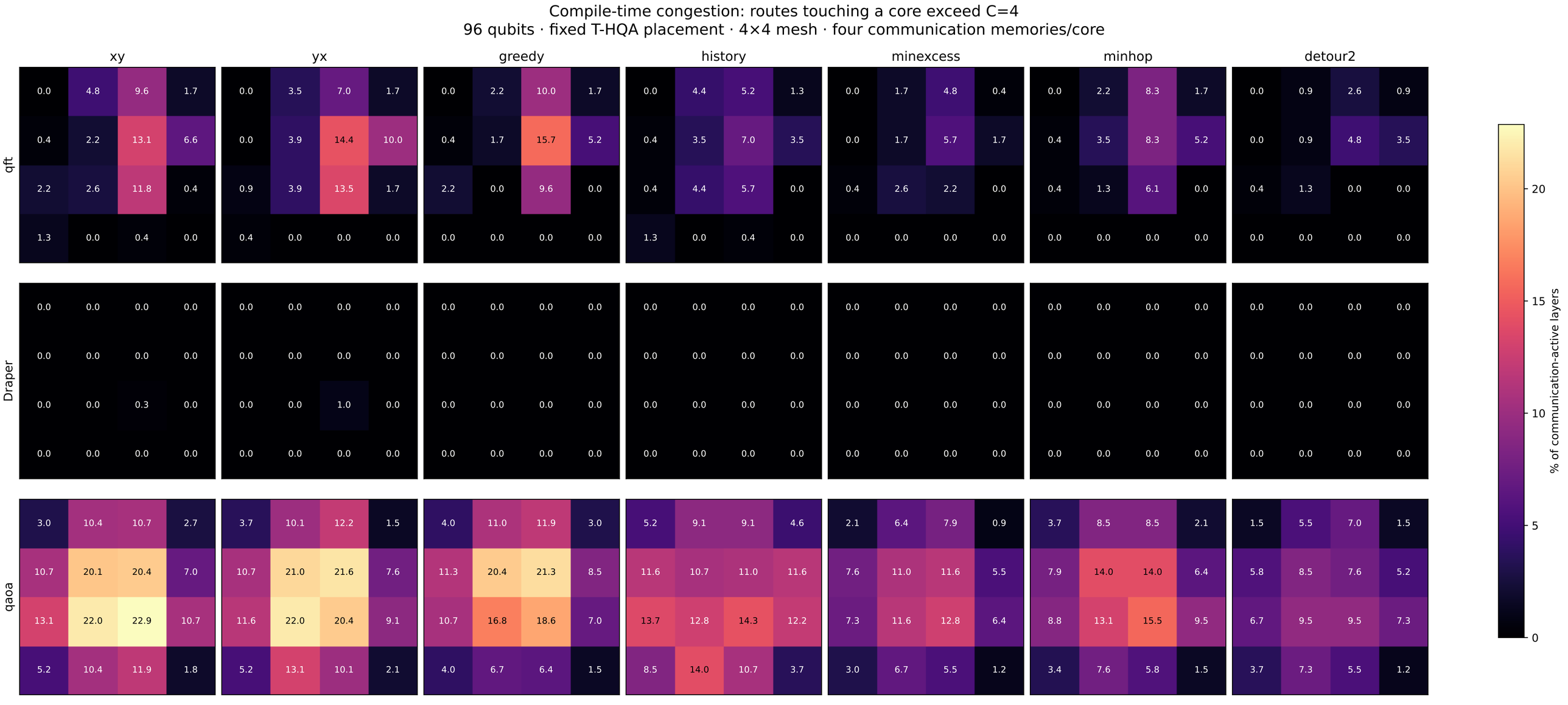
Compile-time congestion: routes touching a core exceed C=4
96 qubits · fixed T-HQA placement · 4×4 mesh · four communication memories/core
xy
yx
greedy
history
minexcess
minhop
detour2
qft
Draper
qaoa
0.0 4.8 9.6 1.7
0.4 2.2 13.1 6.6
2.2 2.6 11.8 0.4
1.3 0.0 0.4 0.0
0.0 3.5 7.0 1.7
0.0 3.9 14.4 10.0
0.9 3.9 13.5 1.7
0.4 0.0 0.0 0.0
0.0 2.2 10.0 1.7
0.4 1.7 15.7 5.2
2.2 0.0 9.6 0.0
0.0 0.0 0.0 0.0
0.0 4.4 5.2 1.3
0.4 3.5 7.0 3.5
0.4 4.4 5.7 0.0
1.3 0.0 0.4 0.0
0.0 1.7 4.8 0.4
0.0 1.7 5.7 1.7
0.4 2.6 2.2 0.0
0.0 0.0 0.0 0.0
0.0 2.2 8.3 1.7
0.4 3.5 8.3 5.2
0.4 1.3 6.1 0.0
0.0 0.0 0.0 0.0
0.0 0.9 2.6 0.9
0.0 0.9 4.8 3.5
0.4 1.3 0.0 0.0
0.0 0.0 0.0 0.0
0.0 0.0 0.0 0.0
0.0 0.0 0.3 0.0
0.0 0.0 0.0 0.0
0.0 0.0 1.0 0.0
3.0 10.4 10.7 2.7
10.7 20.1 20.4 7.0
13.1 22.0 22.9 10.7
5.2 10.4 11.9 1.8
3.7 10.1 12.2 1.5
10.7 21.0 21.6 7.6
11.6 22.0 20.4 9.1
5.2 13.1 10.1 2.1
4.0 11.0 11.9 3.0
11.3 20.4 21.3 8.5
10.7 16.8 18.6 7.0
4.0 6.7 6.4 1.5
5.2 9.1 9.1 4.6
11.6 10.7 11.0 11.6
13.7 12.8 14.3 12.2
8.5 14.0 10.7 3.7
2.1 6.4 7.9 0.9
7.6 11.0 11.6 5.5
7.3 11.6 12.8 6.4
3.0 6.7 5.5 1.2
3.7 8.5 8.5 2.1
7.9 14.0 14.0 6.4
8.8 13.1 15.5 9.5
3.4 7.6 5.8 1.5
1.5 5.5 7.0 1.5
5.8 8.5 7.6 5.2
6.7 9.5 9.5 7.3
3.7 7.3 5.5 1.2
% of communication-active layers
0
5
10
15
20